Reproducing DoRA for Commonsense Reasoning with Student-Scale Compute

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Motivation & Paper Idea

Full fine-tuning large language models is expensive in memory, storage, and training time. **LoRA** reduces this cost by training low-rank adapter updates while freezing the base model, but the low-rank update alone may limit how pretrained weights can change.[1]

DoRA

- **DoRA** decomposes pretrained weights into **magnitude** and **direction**.[3]
- Direction receives a **LoRA**-style low-rank update; magnitude is learned separately.
- The paper reports improved accuracy over **LoRA** across multiple tasks.

Project goal

- Reproduce the paper's LLaMA2-7B commonsense reasoning trend.
- Test whether **DoRA** helps most in attention layers, MLP layers, or only when applied broadly.

Methodology

Model, data, and metrics

- Base model: `meta-llama/Llama-2-7b-hf`.
- Training data: `commonsense_15k.json`, a student-scale commonsense subset.
- Evaluation: BoolQ, PIQA, Social IQA, HellaSwag, WinoGrande, ARC-Easy, ARC-Challenge, OpenBookQA.
- Metrics: per-task accuracy and macro-average accuracy across all eight tasks.

Implementation

- Repo-owned PyTorch **LoRA** and **DoRA** adapter layers.
- Hugging Face `transformers` model/tokenizer loading and generation.
- Frozen base model, adapter-only checkpoints, 3 epochs, cutoff length 256.
- Colab-focused 4-bit NF4 runtime presets for T4, L4, and A100 GPUs.

Independent Scope Ablation

Scope	Target modules	Purpose
Full	`q,k,v,up,down` projections	Paper-level reproduction
Attention only	`q,k,v` projections	Test routing/context layers
MLP only	`up,down` projections	Test feed-forward layers

Design. Each scope compares matched **LoRA** and **DoRA** adapters under the same evaluation pipeline.

Headline Result

77.41%	79.20%	+1.79	6/1/1
Full **LoRA** macro	Full **DoRA** macro	accuracy points	wins/ties/losses

Full-scope **DoRA** reproduced the paper's qualitative improvement over **LoRA** under our student-scale setup.

Macro Accuracy by Adapter Scope

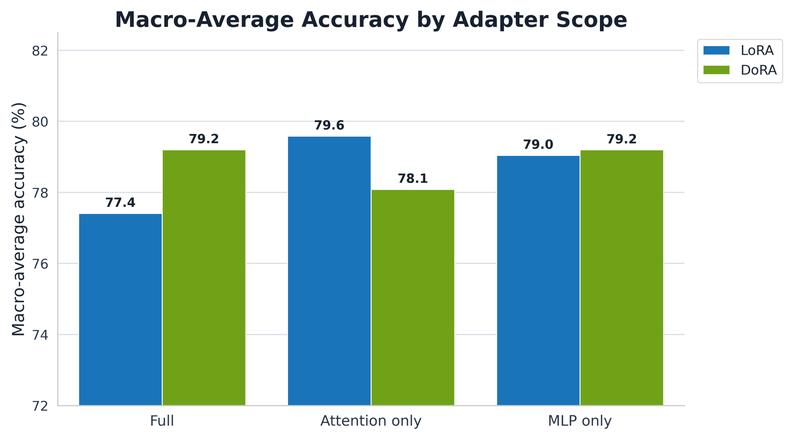

Figure. Full-scope **DoRA** improves macro accuracy from 77.4% to 79.2%. Attention-only **DoRA** does not improve over its matched **LoRA** baseline.

Scope	LoRA macro	DoRA macro	Delta
Full	77.41%	**79.20%**	**+1.79**
Attention only	**79.59%**	78.09%	-1.50
MLP only	79.05%	**79.20%**	+0.15

Where Does DoRA Help?

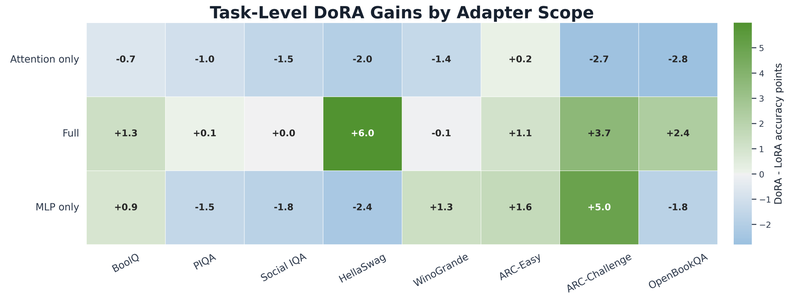

Figure. Task-level **DoRA** minus **LoRA** gains. Full-scope **DoRA** improves 6/8 tasks, ties Social IQA, and slightly loses on WinoGrande.

Largest full-scope gains. HellaSwag +6.0, ARC-Challenge +3.7, OpenBookQA +2.4 accuracy points.

Official Reference Context

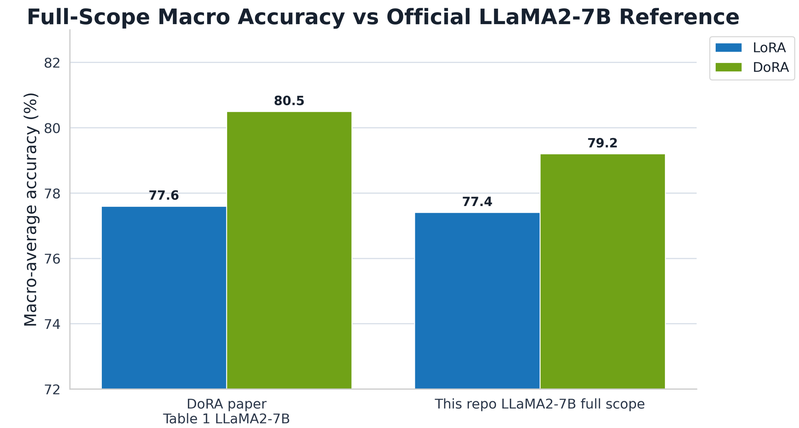

Figure. Our result matches the paper's direction of improvement, but the absolute **DoRA** score remains below the official LLaMA2-7B reference.

Interpretation

- The main paper-level claim **partially reproduced**: full-scope **DoRA** beat full-scope **LoRA**.
- Gains were not uniform across isolated module groups.
- Attention-only **DoRA** underperformed attention-only **LoRA** on 7/8 tasks.
- MLP-only **DoRA** was roughly tied overall: ARC gains were offset by drops on PIQA, Social IQA, HellaSwag, and OpenBookQA.
- Best hypothesis: **DoRA**'s magnitude/direction split works best when attention and MLP transformations can adapt together.

Conclusion & Future Work

Takeaways

- Full-scope **DoRA** improved macro-average accuracy by **+1.79 points**.
- The reproduction supports the qualitative value of weight-decomposed low-rank adaptation.
- The ablation adds a caution: applying **DoRA** narrowly can remove the benefit.

Next steps

- Train on the full `commonsense_170k.json` dataset.
- Repeat all six conditions across multiple random seeds.
- Test Llama-3-8B and larger Llama-family models.
- Add parameter-count, GPU-memory, and wall-clock comparisons.
- Analyze wrong answers by task, especially HellaSwag, ARC-Challenge, and OpenBookQA.

References

[1] Edward J. Hu et al. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022.

[2] Hugging Face. Transformers documentation. `https://huggingface.co/docs/transformers`, 2026.

[3] Shih-Yang Liu et al. DoRA: Weight-decomposed low-rank adaptation. In *Proceedings of the 41st International Conference on Machine Learning*, 2024. Oral presentation.

[4] NVIDIA Research. NVlabs/DoRA: Official implementation. `https://github.com/NVlabs/DoRA`, 2024.